PUSAN NATIONAL UNIVERSITY

Dept. of Aerospace Engineering

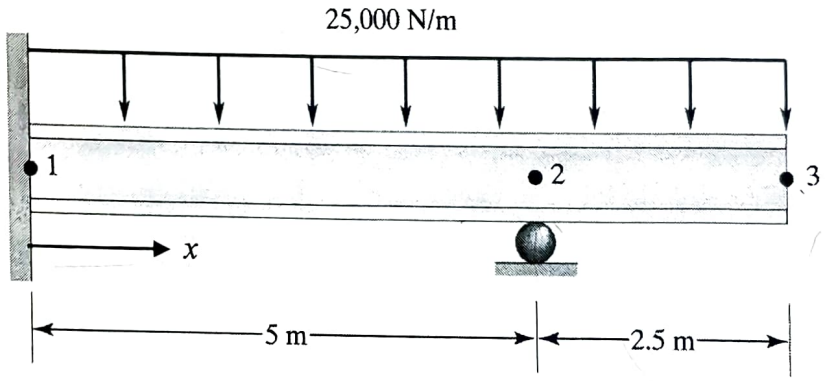
Computational Structural Analysis

Mid-exam. Oct. 23th, 2018

NOTICE: Write down the ANSYS commands on your answer sheet.

Problem 1

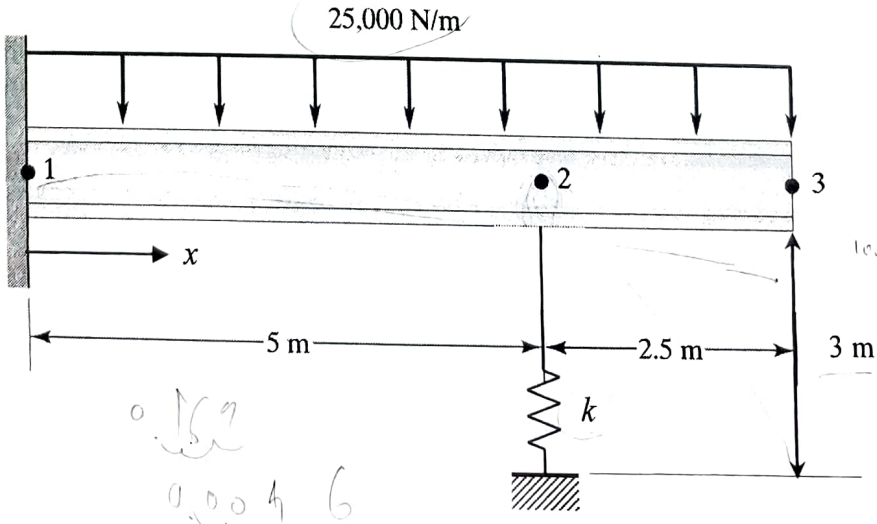
A beam in below figure, whose cross-section is a wide-flange W310 × 52, is made of steel (E = 200 GPa) and supplied from Nucor-Yamato Steel Company. When the beam is subjected to a uniformly distributed load of 25,000 N/m,



(a) Determine the vertical displacement and rotation at x = 7.5m and the rotation at x = 5m.

(b) Check for the validity of FEA results using the static equilibrium conditions.

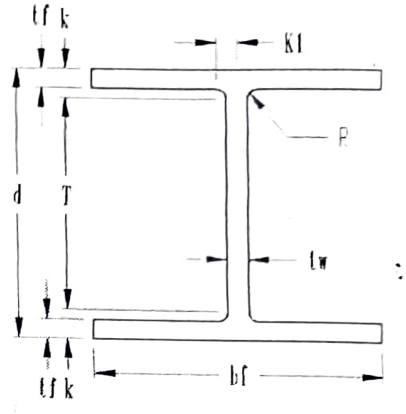
(c) When a linear elastic spring instead of the fixed B.C is attached at x = 5m, as shown in below figure. What is the minimum value of the spring constant k needed to ensure the nodal displacement that is 15mm or less at free end of the beam? Use element type of LINK1 to model the steel spring.



Nucor-Yamato Steel

W310

METRIC



Prime Section Group	Section Size	Area	Depth	Web Thick-ness	Flange Width	Flange Thick-ness	Distance T	Distance k	Distance k_1	Fillet Radius	Elastic Properties X-X I_x	Elastic Properties X-X S_x	Elastic Properties X-X r_x	Elastic Properties Y-Y I_y	Elastic Properties Y-Y S_y	Elastic Properties Y-Y r_y	Plastic Modulus Z_x	Plastic Modulus Z_y
	mm x kg/m	A mm²	d mm	t_w mm	b_f mm	t_f mm	T mm	k mm	k_1 mm	R mm	10^6 mm⁴	10^3 mm³	mm	10^6 mm⁴	10^3 mm³	mm	10^3 mm³	10^3 mm³
310 x 310	W310 x 500	63 700	427	45.1	340	75.1	234	97	43	20	1 690	7 915	163	495	2 900	88.1	9 881	4 490
	W310 x 454	57 800	415	41.3	336	68.7	234	90	41	20	1 478	7 128	160	437	2 606	86.9	8 800	3 998
	W310 x 415	52 800	403	38.9	334	62.7	234	84	39	20	1 294	6 440	156	390	2 343	85.9	7 882	3 605
	W310 x 375	47 800	391	35.4	330	57.2	234	79	38	20	1 132	5 785	154	345	2 081	84.8	7 014	3 212
	W310 x 342	43 700	382	32.6	328	52.6	234	74	36	20	1 007	5 260	152	309	1 885	84.1	6 325	2 900
	W310 x 313	39 900	374	30.0	325	48.3	234	70	35	20	891	4 785	150	276	1 704	83.3	5 703	2 606
	W310 x 283	36 000	365	26.9	322	44.1	234	66	33	20	787	4 310	148	245	1 524	82.6	5 096	2 343
	W310 x 253	32 300	356	24.4	319	39.6	234	61	32	20	687	3 851	146	215	1 349	81.8	4 506	2 065
	W310 x 226	28 800	348	22.1	317	35.6	234	57	31	20	595	3 425	144	189	1 193	81.0	3 982	1 819
	W310 x 202	25 700	341	20.1	315	31.8	234	53	30	20	516	3 048	142	166	1 052	80.3	3 507	1 606
	W310 x 179	22 800	333	18.0	313	28.1	234	50	29	20	445	2 671	140	144	918	79.5	3 048	1 399
	W310 x 158	20 100	327	15.5	310	25.1	234	47	28	20	388	2 376	139	125	808	79.0	2 687	1 231
	W310 x 143	18 200	323	14.0	309	22.9	234	44	27	20	347	2 147	138	112	728	78.5	2 409	1 106
	W310 x 129	16 500	318	13.1	308	20.6	234	42	27	20	308	1 934	137	100	651	78.0	2 163	990
	W310 x 117	15 000	314	11.9	307	18.7	234	40	26	20	276	1 753	136	89.9	587	77.5	1 950	890
	W310 x 107	13 600	311	10.9	306	17.0	234	39	25	20	248	1 596	135	81.2	531	77.2	1 770	806
	W310 x 97	12 300	308	9.9	305	15.4	234	37	25	20	222	1 440	134	72.4	477	76.7	1 586	723
310 x 250	W310 x 86	11 000	310	9.1	254	16.3	236	37	24	19	198	1 278	134	44.5	351	63.8	1 416	533
	W310 x 79	10 100	306	8.8	254	14.6	236	35	23	19	177	1 157	133	39.9	315	63.0	1 277	477
310 x 200	W310 x 74	9 480	310	9.4	205	16.3	236	37	24	19	163	1 052	132	23.4	228	49.8	1 178	349
	W310 x 67	8 520	306	8.5	204	14.6	236	35	23	19	145	946	131	20.8	203	49.5	1 052	311
	W310 x 60	7 610	303	7.5	203	13.1	236	34	23	19	128	844	130	18.4	180	49.3	934	275
310 x 170	W310 x 52	6 650	317	7.6	167	13.2	271	23	12	8.5	119	747	133	10.2	122	39.1	839	188
	W310 x 44.5	5 670	313	6.6	166	11.2	271	21	12	8.5	99	633	132	8.4	102	38.6	706	157
	W310 x 38.7	4 940	310	5.8	165	9.7	271	20	11	8.5	85	547	131	7.2	88	38.4	610	134
310 x 100	W310 x 32.7	4 180	313	6.6	102	10.8	271	21	12	8.5	65	416	125	1.9	38	21.5	480	60
	W310 x 28.3	3 590	309	6.0	102	8.9	271	19	12	8.5	54	349	122	1.6	31	20.9	405	49
	W310 x 23.8	3 040	305	5.6	101	6.7	272	17	11	8.5	43	280	119	1.2	23	19.6	329	37

Problem 2

Create an area as shown in below figure.

